Before vs after: what improved and what it cost

Vroomcheck · effect of adding the ESV26-289 method (Jones et al., *A Check Engine Light for the Automotive Industry*) · results from the Snowflake run

How the comparison was run. One back-test, run four times. The clusters, the 440 detectable recalls (2017 onwards, model year 2015 onwards) and the 24-month window stay fixed; only the signal that raises an alert changes. “Before” is the original Poisson spike test (the Ethan branch). “After” adds the paper's method: **A** the fleet-normalised D/P indicator, **B** the binomial model-year test, **C** event merging.

poisson (before)	a cluster's complaints compared with its own past
paper	the ESV26-289 indicator alone
either	an alert when either signal fires
combined	an alert only when both fire

Improvements

#	Metric	Before	After	Change	Mode	Why it improved	What caused it
1	**Recalls flagged before the recall (detection rate)**	153 / 440 34.8%	254 / 440 57.7%	**+22.9 pts** **+101 recalls (+66%)**	either	The two signals catch different defects. Poisson catches a sudden jump in a cluster's own complaints. The paper's indicator catches a vehicle whose complaint mix is abnormal compared with the rest of the fleet, even when there is no sudden jump (slow-building problems). Together they cover both.	D/P indicator (`indicator.py`) with the 6-month sustained-elevation test, joined to the Poisson flag with OR (`combine_flags`, `VC_DETECTOR=either`)
2	**Project target: ≥ 50% of past recalls flagged early**	✗ 34.8%	✓ 57.7%	target met	either	Same reason as #1: the added signal lifts coverage past the threshold set in the project pitch.	Same as #1
3	**Detection rate with the paper's method alone**	34.8%	44.1%	**+9.3 pts** **+41 recalls**	paper	Before many recalls, the affected vehicle's share of complaints about that component is already above the fleet's for months. The indicator scores every vehicle-component pair, so it also reaches clusters the Poisson test never flagged (1,462 clusters alerted vs 1,217).	Fleet-normalised D/P indicator: Cohen's h between the vehicle's share and the fleet baseline × ln(trailing-year count), mapped to P
4	**Median early-warning lead time**	9.0 mo	17.2 mo (either) 20.2 mo (paper)	**+8 to +11 mo**	either / paper	The indicator works on trailing 12-month counts and reacts to a sustained abnormal share, so it is often already elevated when the 24-month window before the recall opens. The Poisson test waits for a sharp spike, which tends to come later.	Trailing 12-month count window plus the sustained-elevation test. Caveat: lead time is capped at 24 months, so part of the gain reflects long-elevated problems

5	**Alert workload: EMERGING alerts to review**	108	43	**–60%**	**combined**	Requiring both signals keeps only spikes on vehicles that are also abnormal compared with the fleet. It drops spikes caused by model launches, popularity or noise, where the vehicle's complaint mix is normal.	The indicator used as a confirmation gate on the Poisson spike (AND in `combine_flags`, `VC_DETECTOR=combined`)
6	**Alert months per 1,000 cluster-months**	17.2	6.5	**–62%**	**combined**	Same filtering as #5, measured over the whole history rather than today's alerts.	Same as #5
7	**Alerts followed by a matching recall within 24 months (precision proxy)**	94 / 899 10.5%	47 / 403 11.7%	**+1.2 pts**	**combined**	The gate removes slightly more alerts that never led to a recall than alerts that did. The gain is small: many genuine defects are never recalled, so this proxy is a floor, not true precision.	Same as #5
8	**Model-year precision: named years the recall really covers**	0.355	0.540	**+52%**	**B**	Defects usually hit a few model years, not a vehicle line's whole span. The test names only years holding significantly more of the cluster's complaints than an even split, so more of the named years are right.	Binomial test per model year against a uniform baseline, Bonferroni-corrected (α = 0.05 / years in span) (`model_years.py`)
9	**Model years to check per alert**	7.6	2.9	**–62%**	**B**	Only statistically elevated years are reported, which narrows the scope an investigator has to look at.	Same as #8
10	**Model-year overlap (Jaccard)**	0.338	0.362	**+7%**	**B**	Overall the precision gain slightly outweighs the coverage lost (see trade-off #7).	Same as #8
11	**Alerts with named affected model years**	0	1,345 of 1,846 clusters (73%)	**new capability**	**B**	The feature did not exist before. The remaining 27% have complaints spread evenly, so no single year stands out, and the dashboard says so.	Binomial test, shown on each alert in the dashboard and stored in `ML.CLUSTER_MODEL_YEARS`
12	**Popular vehicles flagged just for their volume**	not controlled	controlled	**new capability**	**A**	The indicator compares the vehicle's share of its own complaints with the rest of the fleet, so volume cancels out. Example: the Ford F-150's engine has a P-indicator of –0.9 and is not flagged.	Share-of-own-complaints normalisation, the vehicle left out of its own baseline, and ln(count) damping
13	**Interpretable defect events**	none	2,518 events	**new capability**	**C**	Flagged months close together are joined into one event with a start, an end, a peak and a complaint count, which is easier to review than month-by-month flags.	The paper's event merging: flags less than 12 months apart form one event (`ML.INDICATOR_EVENTS`)

14	**Run-time cost of the paper's method**	—	+0.1 s detect step (local); indicator for the whole fleet 0.6 s local, 13.1 s in Snowflake	**negligible**	**A, B, C**	The indicator is a single vectorised pass (numpy locally, window functions in Snowflake SQL) over about 1.7 million vehicle-component-months.	Vectorised implementation; Snowflake SQL with window functions and a log-factorial table for the exact binomial test
15	**Automated tests**	6 unit tests	11 unit tests + parity check	**+5 tests**	**—**	New tests pin down the behaviour the paper describes; the Snowflake SQL is checked against an independent pandas version (0 flag disagreements over 520,629 rows).	`tests/test_core.py` and the row-by-row parity check

What is not an improvement: the trade-offs

#	Metric	Before	After	Change	Mode	Why it got worse	What caused it
1	**EMERGING alerts to review**	108	433 (paper) 485 (either)	**×4.0 to ×4.5**	**paper / either**	The indicator works at vehicle-component level, so one elevated pair flags every cluster sharing that make, model and top component. Chronic problems stay elevated for years (some events last over 10 years), so they stay on the list.	The indicator's coarse grain (make × model × component) and the paper's long, merged windows
2	**Alert months per 1,000 cluster-months**	17.2	204.2 (paper) 214.9 (either)	**about ×12**	**paper / either**	Same as #1, over the whole history. The paper's test treats overlapping 12-month windows as independent and applies no multiple-testing correction, so it flags readily.	The 6-month t-test on overlapping trailing windows, at 99% confidence, with no correction
3	**Precision proxy (alerts followed by a recall)**	10.5%	10.0% (paper) 10.4% (either)	**flat**	**paper / either**	More alerts, but roughly the same share recalled. The indicator cannot tell a recall-worthy defect from a chronic quality complaint; the paper itself notes that many complaint spikes never become recalls.	The unsupervised indicator: it measures abnormal complaint share, not safety risk
4	**Detection rate in combined mode**	34.8%	13.4%	**–21.4 pts**	**combined**	Both signals rarely fire in the same month: the Poisson test reacts within about 2 months of a surge, while the indicator needs about 5 months to fill its test window.	The AND gate in `combine_flags`, plus the indicator's 6-month window
5	**Median lead over NHTSA opening an investigation**	11.1 mo	0.4 mo (paper) 0.4 mo (either)	**–10.7 mo**	**paper / either**	Measured only for flagged recalls that had an investigation. Many of these investigations opened well before the recall, and the paper's earliest flag inside the 24-month window often falls around the time the investigation opened.	The indicator's timing relative to investigations, combined with the 24-month back-test window
6	**Time to react to a new surge**	≤ 2 months	about 5 months	**about 3 months slower**	**paper**	The sustained-elevation test needs most of its 6-month window to be elevated before it is confident. This is by design: the paper ignores one-off spikes.	The paper's rolling test window (180 days, 6 months here); measured in the unit tests
7	**Model-year coverage: share of the recall's years named**	0.967	0.545	**–44%**	**B**	The binomial test is conservative: it names only clearly over-represented years and misses affected years with fewer complaints. The paper reports the same behaviour (Cobalt 2006–2007 named vs 2005–2010 recalled).	The binomial test with Bonferroni correction

8	**Detect step run time on Snowflake vs local**	4.5 s (local)	16.1–22.1 s (Snowflake)	about ×4	platform	Not caused by the paper: running on Snowflake adds network round trips and uploading results. The paper's method itself adds about 0.1 s.	The move of the ML layer to Snowflake

Cautions to say out loud

Lead time is capped: an alert only counts within the 24 months before the recall, and 20.2 months is close to that cap, so say “flags earlier”, not “always 20 months ahead”.
The precision proxy is a floor: many real defects are never recalled, so “alerts followed by a recall” understates true precision for every detector.
Clusters were built on the full history: this affects before and after equally, so the comparison stays fair.

Headline for the results slide

“The paper's method lets us choose an operating point. With either signal we catch 57.7% of past recalls early, up from 34.8%, which clears our 50% target, with a median warning of about 17 months. With both signals required we cut the review workload by 60%. And the paper's model-year test makes each alert's scope 52% more precise. All of this adds less than a second of compute.”